COMPUTER ENGINEERING

QUEZON CITY

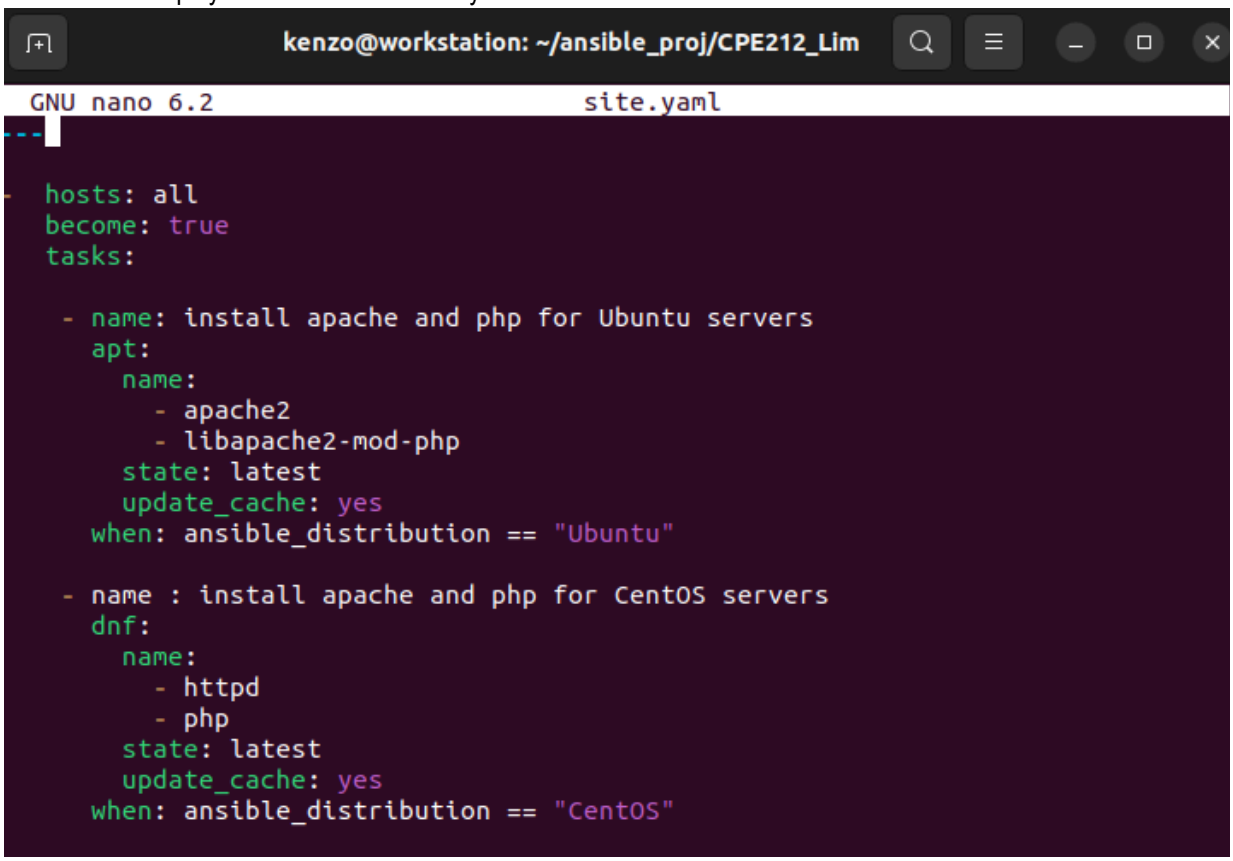
COURSE CODE/TITLE:	CPE212 / Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	Sept 3 2026	DATE SUBMITTED:	Sept 3 2026
NAME:	Kenzo Lim	INSTRUCTOR:	Engr. Jimlord Quejado

ACTIVITY NO. 6.1 **Targeting Specific Nodes and Managing Services**

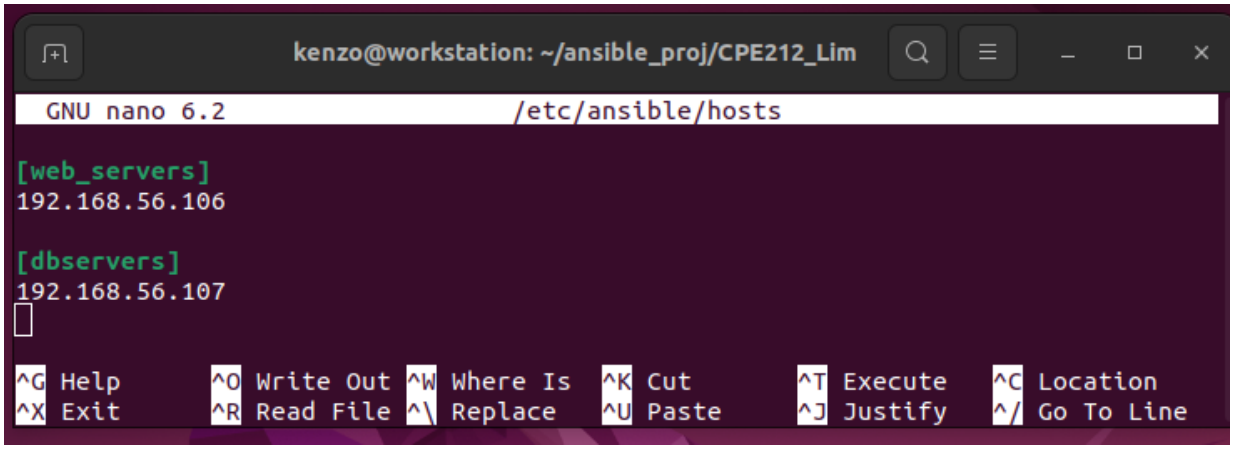
1. PROCEDURE OUTPUT

Task 1: Targeting Specific Nodes

1. Create a new playbook and named it site.yml.

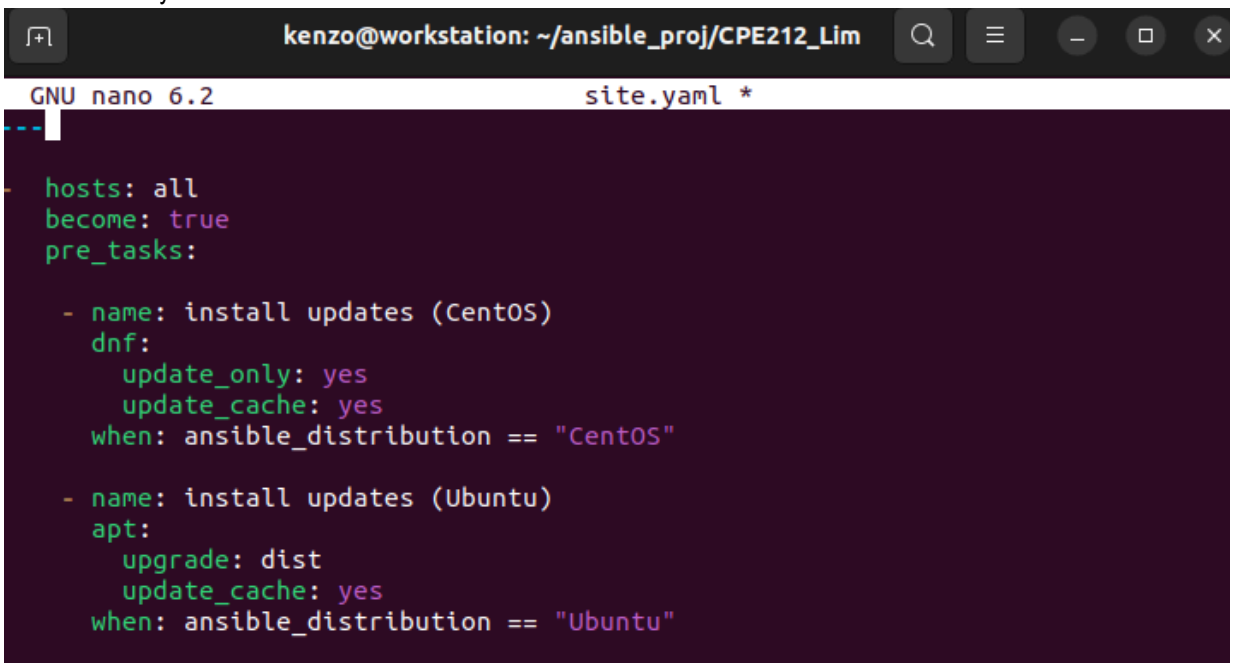


2. Edit the inventory file. Remove the variables we put in our last activity



(in this case, I only have 3 vms running in my personal PC (ubuntu workstation, ubuntu server, centos server))

3. Edit the site.yml

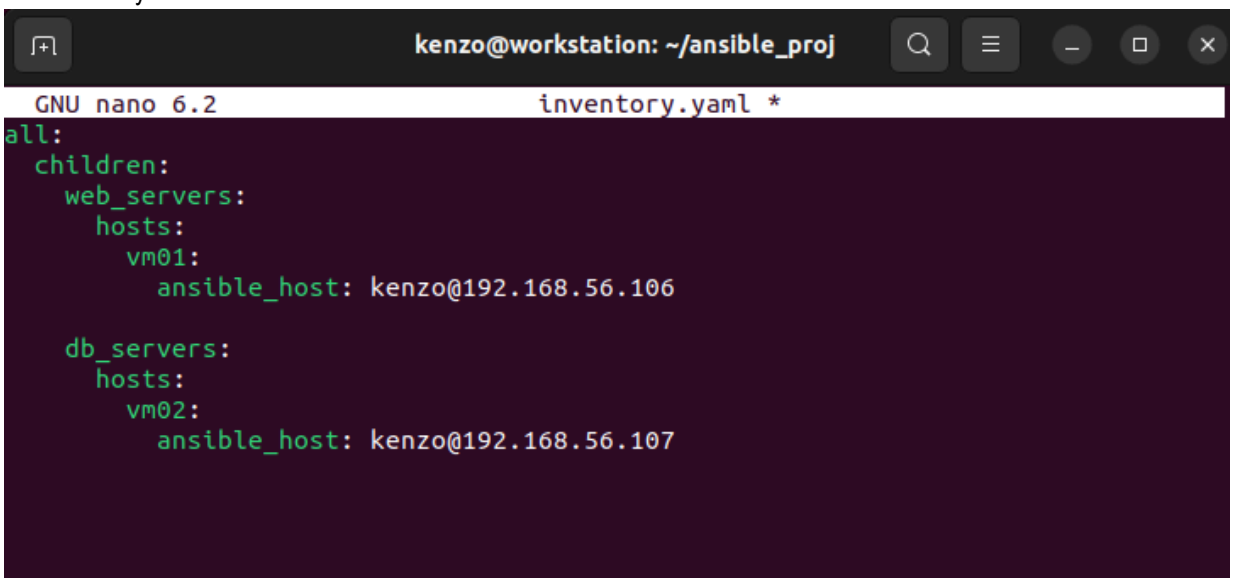


```
-  hosts: web_servers
   become: true
   tasks:

    - name: install apache and php for Ubuntu servers
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name : install apache and php for CentOS servers
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"
```

Run the site.yml file and describe the result



```
kenzo@workstation:~/ansible_proj/CPE212_Lim$ ansible-playbook -i inventory.yaml
site.yaml --ask-become-pass
BECOME password:

PLAY [all] ********************************************************************

TASK [Gathering Facts] ********************************************************
[WARNING]: Platform linux on host vm02 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [vm02]
[WARNING]: Platform linux on host vm01 is using the discovered Python
interpreter at /usr/bin/python3.9, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [vm01]

TASK [install updates (CentOS)] ***********************************************
skipping: [vm02]
ok: [vm01]

TASK [install updates (Ubuntu)] ***********************************************
skipping: [vm01]
```

```
TASK [install updates (CentOS)] ***********************************************
skipping: [vm02]
ok: [vm01]

TASK [install updates (Ubuntu)] ***********************************************
skipping: [vm01]
changed: [vm02]

PLAY [web_servers] ************************************************************

TASK [Gathering Facts] ********************************************************
ok: [vm01]

TASK [install apache and php for Ubuntu servers] ******************************
skipping: [vm01]

TASK [install apache and php for CentOS servers] ******************************
ok: [vm01]

PLAY RECAP ********************************************************************
vm01                       : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0
  ignored=0
vm02                       : ok=2    changed=1    unreachable=0    failed=0    skipped=1    rescued=0
  ignored=0
```

The pre task was able to complete the task on both vms, although on the second task, only vm02 changed which ran apt upgrade: dist while vm01 did not change since it was already updated.

4. Let's try to edit again the site.yml file. This time, we are going to add plays targeting the other servers

```
- name: install mariadb package (CentOS)
  yum:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "CentOS"

- name: "Mariadb- Restarting/Enabling"
  service:
    name: mariadb
    state: restarted
    enabled: true

- name: install mariadb package (Ubuntu)
  apt:
    name: mariadb-server
    state: latest
  when: ansible_distribution == "Ubuntu"
```

Run the site.yml file and describe the result.

```
TASK [Gathering Facts] *********************************************************************
[WARNING]: Platform linux on host vm02 is using the discovered Python interpreter at
/usr/bin/python3.10, but future installation of another Python interpreter could change the meaning
of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [vm02]
[WARNING]: Platform linux on host vm01 is using the discovered Python interpreter at
/usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of
that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [vm01]

TASK [install updates (CentOS)] ************************************************************
skipping: [vm02]
ok: [vm01]

TASK [install updates (Ubuntu)] ************************************************************
skipping: [vm01]
ok: [vm02]

PLAY [web_servers] *************************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [vm01]

TASK [install apache and php for Ubuntu servers] *******************************************
skipping: [vm01]

TASK [install apache and php for CentOS servers] *******************************************
ok: [vm01]

TASK [install mariadb package (CentOS)] ****************************************************
```

It gets stuck in the task installing mariadb package (CentOS)

```
================================================================================
Install  13 Packages

Total size: 19 M
Installed size: 113 M
Is this ok [y/N]: y
Downloading Packages:
[SKIPPED] mariadb-10.5.29-5.el9.x86_64.rpm: Already downloaded
[SKIPPED] mariadb-backup-10.5.29-5.el9.x86_64.rpm: Already downloaded
[SKIPPED] mariadb-common-10.5.29-5.el9.x86_64.rpm: Already downloaded
[SKIPPED] mariadb-connector-c-3.2.6-1.el9.x86_64.rpm: Already downloaded
[SKIPPED] mariadb-connector-c-config-3.2.6-1.el9.noarch.rpm: Already downloaded
[SKIPPED] mariadb-errmsg-10.5.29-5.el9.x86_64.rpm: Already downloaded
[SKIPPED] mariadb-gssapi-server-10.5.29-5.el9.x86_64.rpm: Already downloaded
[SKIPPED] mariadb-server-10.5.29-5.el9.x86_64.rpm: Already downloaded
[SKIPPED] mariadb-server-utils-10.5.29-5.el9.x86_64.rpm: Already downloaded
[SKIPPED] mysql-selinux-1.0.18-1.el9.noarch.rpm: Already downloaded
[SKIPPED] perl-DBD-MariaDB-1.21-17.el9.x86_64.rpm: Already downloaded
[SKIPPED] perl-File-Copy-2.34-483.el9.noarch.rpm: Already downloaded
[SKIPPED] perl-Sys-Hostname-1.23-483.el9.x86_64.rpm: Already downloaded
Running transaction check
Waiting for process with pid 5243 to finish.
```

I tried installing manually, and again it gets stuck waiting for 5243 dnf pid to finish

5. Go to the remote server (Ubuntu) terminal that belongs to the db_servers group and check the status for mariadb installation using the command: systemctl status mariadb. Do this on the CentOS server also.

Since it kept getting stuck at one point, I was not able to download mariadb

```
kenzo@workstation:~$ systemctl status mariadb
Unit mariadb.service could not be found.
```

```
[kenzo@localhost ~]$ sudo systemctl status mariadb
Unit mariadb.service could not be found.
```

6. Edit the site.yml again. This time we will append the code to configure installation on the file_servers group.

```
- hosts: file_servers
  become: true
  tasks:

   - name: install samba package
     package:
       name: samba
       state: latest
```

Since I do not have enough vm for the file_servers, I was not able to run the site.yaml in this part

Task 2: Using Tags in running playbooks

1. Edit the site.yml file. Add tags to the playbook. After the name, we can place the tags: name_of_tag. This is an arbitrary command, which means you can use any name for a tag.

kenzo@workstation: ~/CPE212_Lim

GNU nano 6.2 site.yaml *

```
---

  hosts: all
  become: true
  pre_tasks:

    - name: install updates (CentOS)
      tags: always
      dnf:
        update_only: yes
        update_cache: yes
      when: ansible_distribution == "CentOS"

    - name: install updates (Ubuntu)
      tags: always
      apt:
        upgrade: dist
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

  hosts: web_servers
  become: true
  tasks:

    - name: install apache and php for Ubuntu servers
      tags: apache,apache2,ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: install apache and php for CentOS servers
      tags: apache,centos,httpd
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"

    - name: install mariadb package (CentOS)
      tags: centos, db,mariadb
```

```
    yum:
     name: mariadb-server
     state: latest
    when: ansible_distribution == "CentOS"

  - name: "Mariadb- Restarting/Enabling"
    service:
      name: mariadb
      state: restarted
      enabled: true

  - name: install mariadb package (Ubuntu)
    tags: db, mariadb,ubuntu
    apt:
      name: mariadb-server
      state: latest
    when: ansible_distribution == "Ubuntu"

- hosts: file_servers
  become: true
  tasks:

  - name: install samba package
    tags: samba
    package:
      name: samba
      state: latest
```

2. On the local machine, try to issue the following commands and describe each result:

2.1 ansible-playbook --list-tags site.yml

```
kenzo@workstation:~/CPE212_LLM$ ansible-playbook --list-tags site.yaml

playbook: site.yaml

  play #1 (all): all    TAGS: []
      TASK TAGS: [always]

  play #2 (web_servers): web_servers    TAGS: []
      TASK TAGS: [apache, apache2, centos, db, httpd, mariadb, ubuntu]

  play #3 (file_servers): file_servers  TAGS: []
      TASK TAGS: [samba]
```

It produced an output of the host's tags from play #1 to play #3,

2.2 ansible-playbook --tags centos --ask-become-pass site.yml

```
kenzo@workstation:~/CPE212_Lim$ ansible-playbook --tags centos --ask-become-pass
 site.yaml
BECOME password:

PLAY [all] *********************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.7]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.9 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will
default to using the discovered platform python for this host. See https://docs
.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation
warnings can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.9]
ok: [192.168.56.8]

TASK [install updates (CentOS)] ************************************************
skipping: [192.168.56.8]
skipping: [192.168.56.7]
ok: [192.168.56.9]

TASK [install updates (Ubuntu)] ************************************************
skipping: [192.168.56.9]
changed: [192.168.56.8]
changed: [192.168.56.7]

PLAY [web_servers] *************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.7]

TASK [install apache and php for CentOS servers] *******************************
skipping: [192.168.56.7]

TASK [install mariadb package (CentOS)] ****************************************
skipping: [192.168.56.7]

PLAY [file_servers] ************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.56.9]

PLAY RECAP *********************************************************************
192.168.56.7               : ok=3    changed=1    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.8               : ok=2    changed=1    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

In this part, the tag that was used was centos, meaning it only executed the tasks that were related with the centos tag. As we can see 192.168.56.7 and 192.168.56.8 were changed while other distributions were skipped.

2.3 ansible-playbook --tags db --ask-become-pass site.yml

```
kenzo@workstation:~/CPE212_Lim$ ansible-playbook --tags db --ask-become-pass site.yaml
BECOME password:

PLAY [all] *********************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.7]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.9 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.9]
ok: [192.168.56.8]

TASK [install updates (CentOS)] ************************************************************************************************************
skipping: [192.168.56.7]
skipping: [192.168.56.8]
ok: [192.168.56.9]

TASK [install updates (Ubuntu)] ************************************************************************************************************
skipping: [192.168.56.9]
ok: [192.168.56.8]
ok: [192.168.56.7]

PLAY [web_servers] *************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.7]

TASK [install mariadb package (CentOS)] ****************************************************************************************************
skipping: [192.168.56.7]

TASK [install mariadb package (Ubuntu)] ****************************************************************************************************
changed: [192.168.56.7]

PLAY [file_servers] ************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.9]

PLAY RECAP *********************************************************************************************************************************
192.168.56.7               : ok=4    changed=1    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.8               : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

In this part, the command executes only the tasks marked with the db tag. As we can see, only 192.168.56.7 was changed since it filtered out the non-DB tasks.

2.4 ansible-playbook --tags apache --ask-become-pass site.yml

```
kenzo@workstation:~/CPE212_Lim$ ansible-playbook --tags apache --ask-become-pass site.yaml
BECOME password:

PLAY [all] *********************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.7]
ok: [192.168.56.8]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.9 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.9]

TASK [install updates (CentOS)] ************************************************************************************************************
skipping: [192.168.56.7]
skipping: [192.168.56.8]
ok: [192.168.56.9]

TASK [install updates (Ubuntu)] ************************************************************************************************************
skipping: [192.168.56.9]
ok: [192.168.56.7]
ok: [192.168.56.8]

PLAY [web_servers] *************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.7]

TASK [install apache and php for Ubuntu servers] *******************************************************************************************
ok: [192.168.56.7]

TASK [install apache and php for CentOS servers] *******************************************************************************************
skipping: [192.168.56.7]

PLAY [file_servers] ************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [192.168.56.9]

PLAY RECAP *********************************************************************************************************************************
192.168.56.7               : ok=4    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.8               : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

This part runs only the tasks tagged with apache, the output shows that the playbook skipped all other configurations and focused on the ones tagged with apache

2.5 ansible-playbook --tags “apache,db” --ask-become-pass site.yml

```
TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.56.7]
[DEPRECATION WARNING]: Distribution centos 9 on host 192.168.56.9 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [192.168.56.9]
ok: [192.168.56.8]

TASK [install updates (CentOS)] ************************************************************************************************
skipping: [192.168.56.7]
skipping: [192.168.56.8]
ok: [192.168.56.9]

TASK [install updates (Ubuntu)] ************************************************************************************************
skipping: [192.168.56.9]
ok: [192.168.56.7]
ok: [192.168.56.8]

PLAY [web_servers] *************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.56.7]

TASK [install apache and php for Ubuntu servers] *******************************************************************************
ok: [192.168.56.7]

TASK [install apache and php for CentOS servers] *******************************************************************************
skipping: [192.168.56.7]

TASK [install mariadb package (CentOS)] ****************************************************************************************
skipping: [192.168.56.7]

TASK [install mariadb package (Ubuntu)] ****************************************************************************************
ok: [192.168.56.7]

PLAY [file_servers] ************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [192.168.56.9]

PLAY RECAP *********************************************************************************************************************
192.168.56.7               : ok=5    changed=0    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
192.168.56.8               : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
192.168.56.9               : ok=3    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
```

Again from the previous screenshot this also targeted the ones with apache tags in it, but in this case db was added into the command after apache, making it execute multiple set of tasks.

Task 3: Managing Services

1. Edit the file site.yml and add a play that will automatically start the httpd on CentOS server.

```
  - name: start httpd (CentOS)
    tags: apache, centos, httpd
    service:
      name: httpd
      state: started
    when: ansible_distribution == "CentOS"
```

2. To test it, before you run the saved playbook, go to the CentOS server and stop the currently running httpd using the command sudo systemctl stop httpd. When prompted, enter the sudo password. After that, open the browser and enter the CentOS server’s IP address. You should not be getting a display because we

stopped the httpd service already.



Unable to connect

Firefox can't establish a connection to the server at 192.168.56.9.

- The site could be temporarily unavailable or too busy. Try again in a few moments.
- If you are unable to load any pages, check your computer's network connection.
- If your computer or network is protected by a firewall or proxy, make sure that Firefox is permitted to access the web.

Try Again

3. Go to the local machine and this time, run the site.yml file. Then after running the file, go again to the CentOS server and enter its IP address on the browser. Describe the result.

```
skipping: [server1]
ok: [server3]

TASK [start httpd (CentOS)] ****************************************************************************************************
skipping: [server1]
changed: [server3]

TASK [install mariadb package (CentOS)] ****************************************************************************************
skipping: [server1]
changed: [server3]

TASK [Mariadb- Restarting/Enabling] ********************************************************************************************
changed: [server1]
changed: [server3]

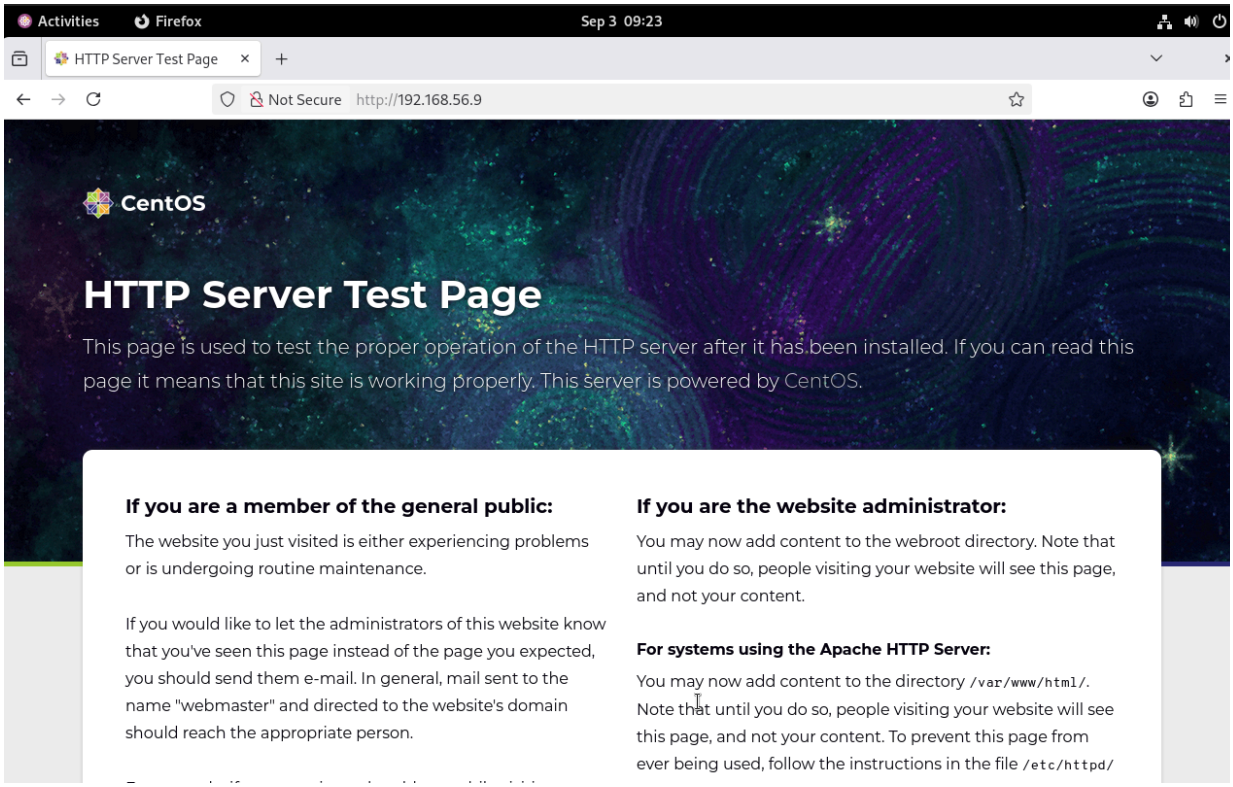
TASK [install mariadb package (Ubuntu)] ****************************************************************************************
skipping: [server3]
ok: [server1]

PLAY [file_servers] ************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [server3]

TASK [install samba package] ***************************************************************************************************
changed: [server3]

PLAY RECAP *********************************************************************************************************************
server1                    : ok=6    changed=1    unreachable=0    failed=0    skipped=4    rescued=0    ignored=0
server2                    : ok=2    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
server3                    : ok=9    changed=4    unreachable=0    failed=0    skipped=3    rescued=0    ignored=0
```



The server went up since the service started.

To automatically enable the service every time we run the playbook, use the command enabled: true similar to Figure 7.1.2 and save the playbook.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Supplementary Activity:
1. Creating a Role for Web Servers
2. Creating a Role for Web Servers
3. Creating a Role for Database Servers
4. Combining Multiple Roles in a Playbook

To orchestrate multiple roles in a single playbook for different server groups.
Add tags to each role to allow selective execution, e.g., tags: web for webserver, tags: db for database.

Using ansible-galaxy role init web and db

```
kenzo@workstation:~/CPE212_Lim$ cd roles
kenzo@workstation:~/CPE212_Lim/roles$ cd web
kenzo@workstation:~/CPE212_Lim/roles/web$ cd tasks
kenzo@workstation:~/CPE212_Lim/roles/web/tasks$ nano main.yml
```

kenzo@workstation: ~/CPE212_Lim/roles/web/tasks

GNU nano 6.2 main.yml

```
---
- name: Install Nginx
  tags:
    - web
    - nginx
  ansible.builtin.apt:
    name: nginx
    state: latest
    update_cache: yes

- name: Start and enable Nginx
  tags:
    - web
    - nginx
  ansible.builtin.service:
    name: nginx
    state: started
    enabled: yes
```

```
kenzo@workstation:~/CPE212_Lim/roles$ cd db
kenzo@workstation:~/CPE212_Lim/roles/db$ cd tasks
kenzo@workstation:~/CPE212_Lim/roles/db/tasks$ nano main.yml
```

kenzo@workstation: ~/CPE212_Lim/roles/db/tasks

GNU nano 6.2 main.yml

```
---
- name: Install MySQL
  tags:
    - db
    - mysql
  ansible.builtin.apt:
    name: mysql-server
    state: latest
    update_cache: yes

- name: Start and enable MySQL
  tags:
    - db
    - mysql
  ansible.builtin.service:
    name: mysql
    state: started
    enabled: yes
```

In this part I configured both of the main.yml file inside the roles.

qknlim HOA6/Supplementary6

Code Blame 17 lines (13 loc) · 179 Bytes

```
1   ---
2
3   - hosts: web_servers
4     become: true
5
6     roles:
7       - role: web
8         tags:
9           - web
10
11  - hosts: db_servers
12    become: true
13
14    roles:
15      - role: db
16        tags:
17          - db
```

This is the main playbook that targets the specific host groups which are web_servers and db_servers, as well as their assigned tags which are web and db.

```
kenzo@workstation:~/CPE212_Lin$ ansible-playbook -i inventory.yaml supplementary6.yaml --ask-become-pass
BECOME password:

PLAY [web_servers] ************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************
ok: [server1]
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is using /usr/bin/python
for backward compatibility with prior Ansible releases. A future Ansible release will default to using the discovered platform
python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [server3]

TASK [web : Install Nginx] ****************************************************************************************************************
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [server3]: FAILED! => {"changed": false, "cmd": "apt-get update", "msg": "[Errno 2] No such file or directory: b'apt-get'", "r
c": 2}
changed: [server1]

TASK [web : Start and enable Nginx] *******************************************************************************************************
fatal: [server1]: FAILED! => {"changed": false, "msg": "Unable to start service nginx: Job for nginx.service failed because the contr
ol process exited with error code.\nSee \"systemctl status nginx.service\" and \"journalctl -xeu nginx.service\" for details.\n"}

PLAY RECAP ********************************************************************************************************************************
server1                    : ok=2    changed=1    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server3                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
```

The first execution of the entire playbook produced an output showing that only Server 1 was changed. The playbook attempted to install Nginx on server 3, but the task failed with an error stating that “apt-get could not be found.” I also noticed that Server 3 (CentOS) was not included in the playbook, meaning that the tasks were only targeting the specified servers and not the CentOS server.

```
kenzo@workstation:~/CPE212_Lim$ ansible-playbook -i inventory.yaml supplementary6.yaml --tags db --ask-become-pass
BECOME password:

PLAY [web_servers] ************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is using /usr/bin/python
for backward compatibility with prior Ansible releases. A future Ansible release will default to using the discovered platform
python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [server3]
ok: [server1]

PLAY [db_servers] *************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************
ok: [server2]

TASK [db : Install MySQL] *****************************************************************************************************************


changed: [server2]

TASK [db : Start and enable MySQL] ********************************************************************************************************
ok: [server2]

PLAY RECAP ********************************************************************************************************************************
server1                    : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server2                    : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
server3                    : ok=1    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

After running the playbook, the output shows that ansible filtered and changed/installed mysql on server 2

```
kenzo@workstation:~/CPE212_Lim$ ansible-playbook -i inventory.yaml supplementary6.yaml --tags nginx --ask-become-pass
BECOME password:

PLAY [web_servers] ************************************************************************************************************************

TASK [Gathering Facts] ********************************************************************************************************************
ok: [server1]
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use /usr/libexec/platform-python, but is using /usr/bin/python
for backward compatibility with prior Ansible releases. A future Ansible release will default to using the discovered platform
python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [server3]

TASK [web : Install Nginx] ****************************************************************************************************************
[WARNING]: Updating cache and auto-installing missing dependency: python3-apt
fatal: [server3]: FAILED! => {"changed": false, "cmd": "apt-get update", "msg": "[Errno 2] No such file or directory: b'apt-get'", "
c": 2}
ok: [server1]

TASK [web : Start and enable Nginx] *******************************************************************************************************
fatal: [server1]: FAILED! => {"changed": false, "msg": "Unable to start service nginx: Job for nginx.service failed because the cont
ol process exited with error code.\nSee \"systemctl status nginx.service\" and \"journalctl -xeu nginx.service\" for details.\n"}

PLAY RECAP ********************************************************************************************************************************
server1                    : ok=2    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server3                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
```

The last execution of the playbook was utilized by using nginx as the tag, again it failed in server 3 since server 3 is CentOS.

Reflections:

Answer the following:

1. What is the importance of putting our remote servers into groups?

Putting remote servers into groups lets us organize them by their specific role, which makes it much easier to manage them together. This also allows us to apply tasks to an entire group at once, without having to configure each server individually.

2. What is the importance of tags in playbooks?

Tags act as labels that let us run only the specific parts of a playbook that we need. You can update or fix a single service without having to run the entire playbook from the beginning.

3. Why do you think some services need to be managed automatically in playbooks?
Automatically managing services makes sure that important applications like web or database servers are always running as they should without needing manual help. This means that services can be configured automatically when needed as it reduces errors and saves time.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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